Physics Informed Diffusion Models

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Outline

1. Recap Diffusion Models
2. Setup: Physics Informed Diffusion Models
3. Training Algorithm
4. Experiments
5. Discussion

Recap Diffusion Models

- First point
- Second point
- Third point

Physics Informed Diffusion Models

Definition

We assume our physics is described by a PDE on $\Omega \subseteq \mathbb{R}^d$ of the form

$$\mathcal{F}\left(u(\xi)\right) = 0 \quad \text{with} \quad \xi \in \Omega.$$

and we have a boundary operator

$$\mathcal{B}\left(u(\xi)\right) = 0 \quad \text{with} \quad \xi \in \partial\Omega.$$

where $u : \Omega \to \mathbb{R}^c$ is the solution of interest. To simplify, we write

$$R(u(\xi)) = [\mathcal{F}(u(\xi)), \mathcal{B}(u(\xi))]^\top$$

.

They only consider

$$[1, ..., n] \times [1, ..., n] \subseteq \mathbb{Z}_n \times \mathbb{Z}_n.$$

Similar to standard image based diffusion models, data is then considered to be given as $x_0 \in \mathbb{R}^{c \times n \times n}$ and $\mathcal{F}, \mathcal{B}$ are discretized accordingly.

Setting up Physics based Training Objective

The residual $\mathcal{R} = 0$ in the non discrete gives us an infinite amount of information. To utilize this, we introduce a loss based on $\mathcal{R}$ via a *virtual observable*.

$$q_{\mathcal{R}}(\cdot \mid x_0) \sim \mathcal{N}(\mathcal{R}(x_0), \sigma^2 I_d)$$

Definition

Define

$$p_\theta(\hat{r}) = \int_\Omega q_{\mathcal{R}}(\hat{r} \mid x_0) p_\theta(x_0) dx_0$$

then we can optimize

$$\arg\max_\theta \mathbb{E}_{\hat{r} \sim \delta_0} \left[\log p_\theta(\hat{r})\right] = \arg\max_\theta \mathbb{E}_{x_0 \sim p_\theta} \left[\log q_{\mathcal{R}}\left(\hat{r} = 0 \mid x_0\right)\right].$$

So we can train our model by evaluating x_0 on the the virtual observable log likelihood. Note that this loss is equivalent to

$$\arg\max_\theta \log p_\theta(0) = \arg\max_\theta \mathbb{E}_{x_0 \sim p_\theta} \left[\log q_{\mathcal{R}}\left(0 \mid x_0\right)\right]$$

We maintain the probabilistic perspective of generative models and introduce the residuals as *virtual observables* (Rixner & Koutsourelakis, 2021) $\hat{\boldsymbol{r}} = \mathbf{0}$, which we consider to be sampled from the distribution

$$q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0) = \mathcal{N}(\hat{\boldsymbol{r}}; \boldsymbol{\mathcal{R}}(\boldsymbol{x}_0), \sigma^2 \boldsymbol{I}). \tag{8}$$

$q_R = p_\theta$?

In the limit $\sigma^2 \to 0$, this recovers the deterministic setting of strictly enforcing the residual equations, as all probability mass is concentrated in $\boldsymbol{\mathcal{R}}(\boldsymbol{x}_0)$. This can be used to compute the virtual likelihood $p_\theta(\hat{\boldsymbol{r}})$, which we expand in terms of drawn samples $\boldsymbol{x}_0$ as

Makesno

$$p_\theta(\hat{\boldsymbol{r}}) = \int p_\theta(\hat{\boldsymbol{r}}, \boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0 = \int q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0) p_\theta(\boldsymbol{x}_0)\, \mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\boldsymbol{x}_0 \sim p_\theta(\boldsymbol{x}_0)} q_{\mathcal{R}}(\hat{\boldsymbol{r}}|\boldsymbol{x}_0). \tag{9}$$

sense

This factorization of $p_\theta(\hat{\boldsymbol{r}}, \boldsymbol{x}_0)$ is reasonable as the residual follows $\boldsymbol{x}_0$ via equation 8. The goal is to maximize the usual log-likelihood over the virtual observable $\hat{\boldsymbol{r}}$, i.e.,

$$\arg\max_\theta \mathbb{E}_{\hat{\boldsymbol{r}}} \left[\log p_\theta(\hat{\boldsymbol{r}})\right] = \arg\max_\theta \mathbb{E}_{\boldsymbol{x}_0 \sim p_\theta(\boldsymbol{x}_0)} \left[\log q_{\mathcal{R}}(\hat{\boldsymbol{r}} = \mathbf{0}|\boldsymbol{x}_0)\right]. \tag{10}$$

define $p_\theta(\hat{r})_\theta = \int_\Omega q_R(\hat{r}|x_0) p_\theta(x_0)\, dx_0$?

Figure

Introducing the Data Loss

We explicitly consider that we have a dataset, which we also want to consider in our training objective. For this, we just additionally introduce the standard model log likelihood as an additional term.

Definition

Our objective is given by

$$\arg\max_{\theta} \mathbb{E}_{x_0 \sim p_\theta} \left[\log p_\theta(x_0)\right] + \arg\max_{\theta} \mathbb{E}_{x_0 \sim p_\theta} \left[\log q_{\mathcal{R}}\left(0 \mid x_0\right)\right]$$

Sample and Mean Estimation

Classical diffusion models try and approximate

$$\hat{x}_0 \approx \mathbb{E}[x_0 \mid x_t]$$

where $\hat{x}_0$ is the model. Then, use $\hat{x}_0$ directly for the loss. This is called mean estimation. However, if we just compute our physics loss via plugging the model output into the residual we have

$$\mathcal{R}(\hat{x}_0) \approx \mathcal{R}(\mathbb{E}[x_0 \mid x_t]) \neq \mathbb{E}[\mathcal{R}(x_0 \mid x_t)]$$

which would be the correct expected residual. To mitigate this, they increase the variance as t increases. Another solution, which is more costly during training, is to use DDIM sampling, which will yield us a sample x_0^* of p_θ, which yields us a better estimate for the residual, this they call sample estimation.

Full Training Algorithm

Algorithm Physics-informed diffusion model training

Set $\bar{\Sigma}_t = \Sigma_t / c = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t} \cdot \frac{\beta_t}{c} I$
repeat
    Sample $x_0 \sim q(x_0), t \sim \text{Uniform}\{1, \ldots, T\}, \epsilon \sim \mathcal{N}(0, I)$
    Compute

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$$

    Estimate x_0^* via $\hat{x}_0 \approx \mathbb{E}[x_0 \mid x_t]$ or DDIM sampling.
    Take a gradient descent step on

$$\nabla_\theta \left[\lambda_t \left\| x_0 - \hat{x}_0(x_t, t) \right\|^2 + \frac{1}{2\Sigma_t} \left\| \mathcal{R}(x_0^*(x_t, t)) \right\|^2 \right]$$

until converged or for fixed number of iterations

Experimental Setup

They consider only one set of PDEs, namely the Darcy Flow equations given by

$$u(x) = -K(x)\nabla p(x)$$
$$-\nabla \cdot u(x) = f(x)$$

where K is the permeability field, p the pressure field and f a source term. They describe fluid flow through porus media e.g. materials with small holes like sponges.

Experimental Results

Discussion

- They only consider 2d data and PDEs on $\mathbb{R}^2$, how this works on higher dimensions is not tested.